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B. Tech Degree VI Semester (Supplementary) Examination, September 2008

ME 605 CONTROL SYSTEM ENGINEERING

(1999 Scheme)

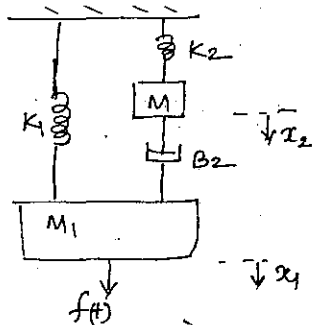
Time : 3 Hours

Maximum Marks : 100

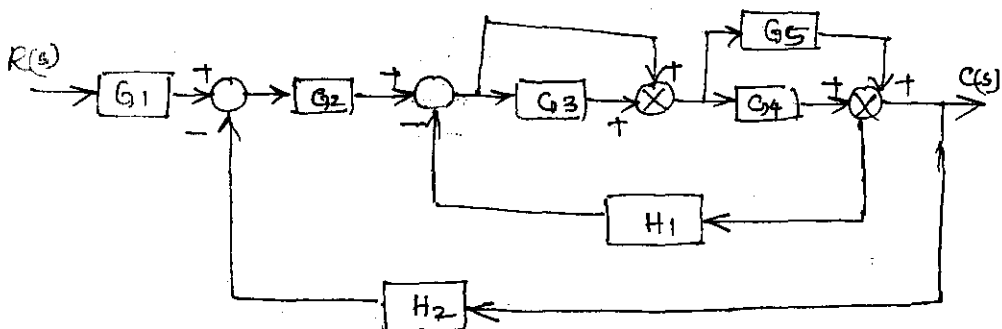
- I. (a) Distinguish between open loop and closed loop systems with examples. (8)
 (b) State and explain initial value theorem and final value theorem. (4)
 (c) Obtain the inverse laplace transform of $F(s) = \frac{5(s+2)}{s^2(s+1)(s+3)}$. (8)

OR

- II. (a) Solve the following differential equation $x'' + 2wnx' + w_n^2x = 0$ where $x(0) = a, x'(0) = b$ where a and b are constants. (10)
 (b) Obtain the mathematical model of the mechanical system shown in the figure. Obtain the closed loop transfer function $x_1(s)/F(s)$. (10)



- III. (a) State Masons Gain formula for the closed loop control-ratio of a multiple loop system, clearly defining all the terms. (6)
 (b) Simplify the block diagram in the figure given below, draw the signal flow graph and verify the same using Masons formula. (14)

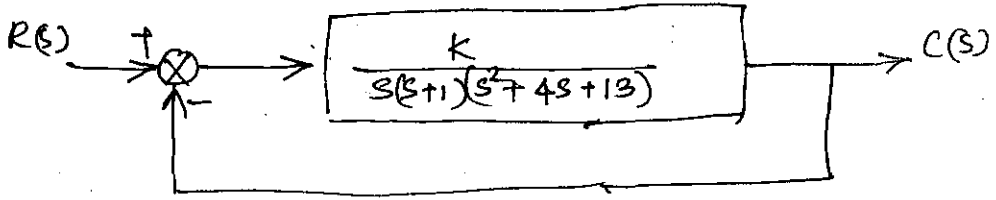


OR

- IV. (a) For a second order system having $M(s) = \frac{C(s)}{R(s)} = \frac{Wn^2}{s^2 + 2\zeta Wn s + Wn^2}$ derive the response $c(t)$ for a unit step input. (10)
 (b) Derive the expressions for peak overshoot M_p , and peak time t_p of the time response of a second order system subjected to a exact step input. (10)

(Turn Over)

- V. (a) State and explain Routh- Hurwitz criterion. How is this used to determine the closed loop stability of control system? (8)
- (b) Determine the range of K for which the system shown in the figure is stable. (12)



OR

- VI. (a) Explain Gain Margin and Phase Margin. (6)
- (b) Sketch the Bode plot for the following transfer function and determine the
- | | |
|--------------------------------|--------------------------------|
| (i) Phase cross over frequency | (ii) Gain cross over frequency |
| (iii) Phase Margin | (iv) Gain Margin |
| (v) Stability of the system | |

$$G(s)H(s) = \frac{200(s+2)}{s(s^2+10s+100)} \quad (14)$$

- VII. (a) Obtain the state space model of the system whose transfer function is given below and draw the block diagram and SFG.

$$\frac{Y(s)}{U(s)} = \frac{s^2 + 3s + 3}{s^3 + 2s^2 + 3s + 1} \quad (10)$$

- (b) Obtain inverse z transform of the following terms.

$$(i) \quad Y(z) = \frac{1}{(1+z^{-1})(1-2^{-2})^2} \quad \text{ROC } |z| > 1$$

$$(ii) \quad Y(z) = \frac{1}{z^2 - 1.2z + 0.2} \quad \text{ROC } |z| > 1. \quad (10)$$

OR

- VIII. (a) A system is governed by the differential equation

$$3\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 10y = 6u(t). \quad \text{Obtain the state space representation.}$$

Draw the block diagram and signal flow graph. (10)

- (b) Obtain the z transform of the following sequences

$$(i) \quad x(n) = n(-1)^n u(n)$$

$$(ii) \quad x(n) = \left(\frac{1}{2}\right)^{n-1} u(n-1). \quad (10)$$

- IX. (a) Derive the transfer function of an armature controlled dc motor. (10)
- (b) Write short notes on

(i) Stepper motor

(ii) Synchros

OR

- X. Write short notes on :

(i) Pneumatic controller

(ii) Hydraulic controller

(iii) Stepper motor

(iv) Tachogenerator. (20)